

The Ouroboros Thesis

Axiom-Free Conditional Uniqueness of the Lutar Invariant:
A Machine-Verified Trust Foundation for Governed Agentic AI

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Abstract

Governed deployment of agentic artificial intelligence in regulated and defense settings requires more than benchmark accuracy: it requires *checkable* guarantees about what a system did, why it was permitted to do it, and whether the record can be tampered with after the fact. This is **v24** of the Ouroboros thesis program of SZL Holdings, a DOI-pinned lineage (v1–v23) whose subject is a machine-verified trust substrate resting on three pillars: (i) the *Ouroboros loop*, a bounded, well-founded self-governing computation in which “the loop is the product”; (ii) the *Lutar invariant* $\Lambda_k(x) = (\prod_{i=1}^k x_i)^{1/k}$, an equal-weight geometric-mean trust aggregator; and (iii) a *proof-trail / receipt architecture* formalized in Lean 4. The headline advance of v24 is an **axiom-free conditional Λ -uniqueness theorem**. In v23 the conditional uniqueness of Λ was gated on a *declared* project axiom (A6’ block-consistency). Here we report **Lutar.Round13.lambda_unique_of_separable**: under {A1, A2, A3, A5} together with *slice-multiplicativity* (separability), Λ is the unique aggregator, with `#print axioms = {propext, Classical.choice, Quot.sound}`— i.e. **no project axiom**, kernel-clean, CI-green, present at `main ba1050b7` (originally merged at `b910c276`). This strictly strengthens v23: the extra hypothesis is now a *checkable property of Φ* (weaker than the old *Factors* premise) discharged through Mathlib and already-proved in-tree lemmas, rather than a declared idealization. The further advance reported in this revision (**v24.1**) is that **CUT-1 — the Aczél quasi-arithmetic representation theorem — is now fully closed on its stated, checkable hypotheses (§4.7)**. Across Waves 18–22 the forward representation is constructed end to end: a Burai–Kiss–Szokol (BKS) density engine, the analytic generator-collapse, the monotone-extension uncountability step, and — newly in Wave 22 — the BKS Fourth-step (*C-order*) *gap-shift ordering*, which is now *derived* from the quasi-arithmetic structure rather than assumed. The net effect is a *sharpened* conditional Λ -uniqueness theorem, `cut1_sharp_conditional_lambda`, that drops two hypotheses (slice-bisymmetry and unit-normalization, both shown redundant). **This sharpens the conditional result; it does *not* make Λ unconditional**. Crucially, the *unconditional* uniqueness of Λ under {A1–A5} remains machine-checked **false** (in-tree `maxAgg_ne_Lambda`; `maxAgg` and `min` are A1–A5 counterexamples), so Λ stays **Conjecture 1** unconditionally — never a theorem. We also report a separate *experimental*, *CI-green* tier of ~ 185 kernel-clean frontier theorems (Waves 11–22: CF-13, CF-17, the Wave-13 replay/quorum/HM results, the Wave-14 pack CF-18/19/20/21, the Wave-15 CF-22 simplex-conditional DPO repair, the Wave-17 CF-23 full binary Pinsker inequality, CF-24–28, and the Wave-18–22 CUT-1 construction), all `#print axioms` -clean, that is never folded into the locked-five. The honesty doctrine is binding throughout: locked-proven = exactly five {F1, F11, F12, F18, F19}; declared axioms disclosed;

supply-chain posture SLSA L1 honest with an L2 build-attestation present and L2-verified / L3 / FedRAMP a roadmap (correcting a v23 overstatement); nothing fabricated. We argue that replacing a declared axiom with a checkable property is an *epistemic upgrade*, and that this calibrated honesty is what makes every positive claim credible to an auditor.

Keywords: governed agentic AI, trust aggregation, Lutar invariant, geometric mean, slice-multiplicativity, functional-equation uniqueness, Lean 4, Mathlib, axiom-free verification, `#print axioms`, receipt chain, SLSA, Byzantine consensus, philosophy of mathematics.

Honesty note (verbatim). The Lutar invariant Λ is **Conjecture 1** unconditionally and is *never* claimed proven unconditionally; unconditional uniqueness under A1–A5 is machine-checked *false*. The v24 conditional uniqueness theorem `lambda_unique_of_separable` holds under $\{A1, A2, A3, A5\} + \text{slice-multiplicativity}$ with *no project axiom* (`#print axioms = {propept, Classical.choice, Quot.sound}`). **CUT-1 is closed on its stated hypotheses** (the quasi-arithmetic *representation* theorem is complete and *sharpens* the conditional Λ result); this does *not* make Λ unconditional — the *unconditional* uniqueness of Λ stays Conjecture 1 and is machine-checked false. Locked/proven = exactly five formulas $\{F1, F11, F12, F18, F19\}$ at `c7c0ba17`. The experimental CI-green tier (~ 185 kernel-clean theorems across Waves 11–22; drift baseline 1323 decl / 23 axioms / 307 sorries at `main ba1050b7`) is a *separate* tier and is never folded into the locked-five. SLSA **L1 honest, L2 build-attestation present**; *not* L2-verified, L3, FedRAMP, Iron Bank, or CMMC. No fabricated results; no fake citations; 0 runtime CDN.

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1 Introduction

Agentic AI systems — models that plan, call tools, retrieve, and act over many steps — are entering domains where the cost of an undetected error is not a bad recommendation but a regulatory breach, a safety incident, or a compromised mission. In such regulated and defense settings the operative question is no longer “how accurate is the model on a benchmark?” but “can the operator *prove*, after the fact and to a skeptical third party, that an action was authorized, that the authorization rested on checkable evidence, and that the record has not been altered?” This is the problem of *governed AI*: an execution substrate in which trust is *earned by checkable evidence* rather than asserted by a vendor. The verifiable-claims program argues that the industry default — assert safety, keep the evaluation private, log to a mutable store — is structurally inadequate [7], and policy bodies have echoed the demand that developers be able to *prove* properties of their systems [22].

This paper is **v24** of the Ouroboros thesis program, a DOI-pinned lineage of twenty-three prior versions (v1–v23) archived under a Zenodo concept record [20]. Each prior version advanced a piece of the substrate; v23 “Unified Substrate” consolidated them under an *honesty doctrine* (§2) that distinguishes, at every point of assertion, *proven* (kernel-verified, Lean-core axioms only) from *proven under a declared axiom* from *conjectured*.

The v24 advance, stated precisely. v23’s strongest Λ -uniqueness result was *conditional* and *gated on a declared project axiom*: `lambda_unique_under_block`, which assumed a named `A6'_block_consistent` idealization. The headline of v24 is that this gate can be removed. The in-tree theorem

`Lutar.Round13.lambda_unique_of_separable`

proves Λ -uniqueness *conditional* on $\{A1, A2, A3, A5\}$ together with *slice-multiplicativity* (separability), and its trusted base is exactly the Lean core: `#print axioms = {propext, Classical.choice, Quot.sound}`— **no project axiom**. It is kernel-clean, CI-green, and merged to `main`. This *strictly strengthens* v23: the extra hypothesis is now a *checkable structural property* of the candidate aggregator Φ — and one weaker than the old *Factors* premise — discharged through Mathlib and already-proved in-tree lemmas, rather than a disclosed-but-unprovable idealization. We characterize this as an *epistemic upgrade* (§9).

The v24.1 advance: CUT-1 closed on its stated hypotheses. This revision (v24.1) refreshes v24 to the *current* proof state at `main ba1050b7` (Waves 11–22 all merged, CI-green). The headline of v24.1 is that **CUT-1 — the Aczél quasi-arithmetic *representation* theorem — is now fully closed on its stated, checkable hypotheses** (§4.7). Across Waves 18–22 every step of the BKS forward construction is discharged kernel-clean, ending with the Wave-22 derivation of the BKS Fourth-step ordering. The payoff is a *sharpened* conditional Λ -uniqueness theorem, `cut1_sharp_conditional_lambda`, on the weakest checkable hypothesis set $\{A1\text{--}A5\}$ + separability + slice-multiplicativity + slice-monotonicity. We state this precisely because it is easy to overclaim: **CUT-1 closed on its stated hypotheses sharpens the *conditional* Λ result; it does *not* make Λ unconditional**. The unconditional conjecture stays open — and stays machine-checked false (next paragraph).

What does *not* change. The single most important honest statement in this program is unchanged and we repeat it without softening: the *unconditional* uniqueness of Λ under $\{A1\text{--}A5\}$ is

machine-checked **false**. The in-tree witness `maxAgg_ne_Lambda` exhibits concrete A1–A5 aggregators (`maxAgg` and `min`) that differ from Λ . We therefore label Λ **Conjecture 1** unconditionally — never a theorem — exactly as v23 did. v24 does not, and never will, claim otherwise.

The frontier tier. Alongside the headline theorem, v24.1 reports ~ 185 *experimental, axiom-free, CI-green* frontier theorems across Waves 11–22 (§6): CF-13 (a deep-equilibrium input-Lipschitz well-posedness margin [5]), CF-17 (a floating-point summation error bound [16]), the Wave-13 results (replay-root completeness, a non-Byzantine quorum shadow, an HM-bottleneck lemma), the Wave-14 pack — CF-18 (Mādhava/Leibniz alternating-series remainder [25]), CF-19 (a Reed–Solomon MDS distance lower bound [26, 27]), CF-20 (VCG efficiency + truthfulness core [29, 10, 15, 23]), and CF-21 (the Cover–Thomas log-sum + Gibbs inequalities [11]) — the Wave-15 CF-22 (a simplex-conditional repair of the false-as-stated DPO KL axiom), the Wave-17 CF-23 (the *full binary Pinsker inequality*), and CF-24–28 (geoBin Aczél axioms, Λ scale-invariance, the abacus place-value lemma, monotone-DEQ unique equilibrium, recurrent-depth Lipschitz contraction), culminating in the Wave-18–22 CUT-1 construction (§4.7). Every one is `#print axioms` -clean. None is folded into the locked-five.

Reading guide. §2 states the honesty doctrine. §3 recalls the Ouroboros loop. §4 treats Λ and its uniqueness boundary, including the v24 axiom-free theorem (§4.4) with its in-tree lemma chain and worked examples, and the v24.1 **CUT-1 closure** on its stated hypotheses (§4.7). §5 covers receipts. §6 documents the locked kernel and the frontier families with signatures and axiom-cleanliness. §7 gives the detailed verification status (Waves 11–22, the drift gate, `#print axioms` discipline). §8–§9 make the honest and philosophical case. §10–§11 state the empirical posture and the limitations. Appendices give reproduction and a glossary.

2 The honesty doctrine

The doctrine is load-bearing and is preserved verbatim across all SZL artifacts. It defines tiers of epistemic status; every claim in this paper carries exactly one tag.

[locked / kernel-verified] Proven in Lean 4 with only the trusted core axioms `{propext, Classical.choice, Quot.sound}`; sorry-free; pinned at a fixed commit and part of the *locked* count. Locked-proven = exactly five.

[experimental, axiom-free, CI-green] Proven sorry-free with only the trusted core axioms, CI-green at the recorded commit, but residing in the *experimental* tier and **never folded into the locked count**.

[axiom-gated] Sorry-free *given* an explicitly declared, disclosed idealizing axiom (e.g. hash collision-resistance), with the axiom appearing in the `#print axioms` ledger.

[CI-pending] Signature-checked but not reproducibly green in the wired build; **not** claimed proven.

[machine-checked FALSE] The statement is refuted by a machine-checked counterexample.

[Conjecture 1 – NOT a theorem] A research hypothesis. Λ ’s unconditional uniqueness lives here.

Non-negotiable rules.

1. **Λ is Conjecture 1 unconditionally** — never claimed proven unconditionally. Unconditional uniqueness under A1–A5 is machine-checked false (`maxAgg_ne_Lambda`). Conditional theorems are claimed only with their hypotheses stated.
2. **Locked-proven = exactly five** {F1, F11, F12, F18, F19} at `c7c0ba17` (749 decl / 14 unique axioms / 163 sorries). This number is never inflated.
3. **The experimental CI-green tier is separate.** At main `ba1050b7` (Waves 11–22 merged) the tier comprises ~ 185 kernel-clean theorems; the drift-gated baseline is 1323 declarations / 23 axioms (22 unique) / 307 sorries (the Wave-15–22 work is additive, registered under `EXPERIMENTAL_SCOPES`, so the locked baseline is unmoved). Nothing in this tier is ever folded into the locked-five.
4. **All axioms are disclosed** via `#print axioms`. No fabricated results; no fake citations.
5. **Supply-chain posture: SLSA L1 honest, L2 build-attestation present.** We do *not* write “L2-verified”, “L3”, “FedRAMP”, “Iron Bank”, or “CMMC”; those remain roadmap. (§7.4 corrects a v23 overstatement.)
6. **Open sorries are disclosed** (§11); 0 runtime CDN (vendored libraries in-image; only live *data* fetches to public APIs).

3 The Ouroboros loop: bounded self-governance

The first pillar is a self-referential governed computation — the *Ouroboros loop* — in which the system’s output feeds its own governance input under a bounded, well-founded recursion: perceive, propose, gate, act, record, and feed the record back. “The loop is the product”: the governed recurrence, with its receipts, is the unit of value, not a single inference.

Definition 3.1 (Bounded Ouroboros loop). A loop is *bounded* if it admits a well-founded measure — a ranking function into a well-ordered set that strictly decreases on each governed step, with no infinite descending chain.

Termination is thereby guaranteed; self-reference is rendered coherent by *Tarskian stratification* (a governed object-level computation is audited by a strictly higher meta-level), avoiding the self-verification paradoxes of Gödel and Tarski. The substrate does **not** claim to prove its own consistency; instead proofs are checked by the Lean kernel, the kernel’s run by continuous integration, and CI by human sign-off. The bottom turtle — the Lean kernel and its core axioms, the declared cryptographic idealizations, and human sign-off — is finite and disclosed (§A.3).

4 The Lutar invariant Λ and its uniqueness boundary

4.1 Definition and the aggregation axioms

A governed action accumulates evidence along several axes (provenance, policy-compliance, measurability-honesty, moral-grounding, ...), each a sub-score in $[0, 1]$. The governance question is how to combine k sub-scores into one trust verdict.

Definition 4.1 (Lutar invariant). For $x = (x_1, \dots, x_k) \in [0, 1]^k$, the *Lutar invariant* is the equal-weight geometric mean

$$\Lambda_k(x) = \left(\prod_{i=1}^k x_i \right)^{1/k}.$$

The geometric mean is the natural aggregator for *ratio-scale* evidence: it commutes with a common rescaling of all axes, and a single zero axis drives the verdict to zero — a disqualifying failure cannot be averaged away. The governance axioms on a candidate aggregator $\Phi : [0, 1]^k \rightarrow [0, 1]$ are:

Axiom 1 (Monotonicity, A1). Φ is non-decreasing in each argument.

Axiom 2 (Positive homogeneity, A2). $\Phi(c \cdot x) = c \Phi(x)$ for $c > 0$ (scale-covariance).

Axiom 3 (Idempotence / normalization, A3). $\Phi(c, \dots, c) = c$.

Axiom 4 (Boundedness, A4). $\Phi(x) \leq \max_i x_i$.

Axiom 5 (Permutation invariance, A5). $\Phi(x_{\sigma(1)}, \dots, x_{\sigma(k)}) = \Phi(x)$ for every permutation σ .

The classical theory (Kolmogorov 1930 [18], Aczél 1948 [1], Hardy–Littlewood–Pólya 1934) confirms that A1–A4 (and even A1–A5) *alone* do not force the geometric mean: the missing ingredient is a *bisymmetry* / *associativity* / *replacement* family [30, 2].

4.2 The unconditional refutation: FALSE, machine-checked

It is tempting to claim Λ_k is the *unique* Φ satisfying A1–A5. **This is false, and we prove it false.**

Theorem 4.2 (Refutation of unconditional uniqueness). [MACHINE-CHECKED FALSE] [CI-GREEN] *There exists $\Phi \neq \Lambda_k$ satisfying A1–A5. In particular the max-aggregator $\max\text{Agg}(x) = \max_i x_i$ satisfies A1–A5 yet differs from Λ_k : at $(4, 1)$ (on the ratio scale) one has $\max\text{Agg} = 4$ while $\Lambda_2(4, 1) = 2$. The minimum aggregator $\min_i x_i$ is a second witness.*

Lean reference. The in-tree witness is `Round13.maxAgg_ne_Lambda`. The max function is monotone (A1), positively homogeneous (A2), idempotent (A3), bounded by itself hence $\leq \max$ (A4), and symmetric (A5); evaluation at $(4, 1)$ separates it from the geometric mean by `decide`. `#print axioms maxAgg_ne_Lambda` reports Lean-core axioms only. `min` is an analogous `decide-checked` companion. $\square$

This is the epistemic heart of the program: we did not merely *fail* to prove unconditional uniqueness; we *proved its negation*. Consequently Λ remains Conjecture 1 unconditionally (§4.5).

4.3 The old route: uniqueness given factorization, and the A6' gate

The maximal honestly-true uniqueness statement of the prior lineage was conditional on *factorization*, fully proved in-tree.

Theorem 4.3 (Uniqueness given factorization). [EXPERIMENTAL, AXIOM-FREE, CI-GREEN] *Let Φ satisfy A1–A5 and suppose Φ factors: there exist exponents $\alpha_1, \dots, \alpha_k \geq 0$ with $\Phi(x) = \prod_i x_i^{\alpha_i}$ for all x . Then $\Phi = \Lambda_k$.*

Lean reference and sketch. The Lean term is `lambda_unique_of_factors` (Round-13). Given factorization, idempotence (A3) forces $\sum_i \alpha_i = 1$ and symmetry (A5) forces all α_i equal, whence $\alpha_i = 1/k$ and $\Phi = \Lambda_k$. The exponent collapse uses `NNReal.rpow` arithmetic and a `Finset` induction; the companion `lambda_factors` (axiom-free, CI-green) shows Λ_k itself factors with exponents $1/k$, so the hypothesis is non-vacuous. $\square$

In v23 the substrate’s *recommended* route turned the opaque factorization hypothesis into a governance-legible *declared axiom*.

Axiom 6 (Block-consistency / aggregation-invariance, A6′). Aggregating evidence within independent blocks and then across the block results equals aggregating the flattened collection; equivalently, the verdict is invariant to how the auditor partitions evidence into review blocks (after Csátó 2018 [12]).

Theorem 4.4 (v23 conditional uniqueness under declared A6′). [AXIOM-GATED] [CI-GREEN] Under $\{A1-A5\}$ together with the single declared axiom *A6′_block_consistent*, Λ_k is the unique normalized aggregator. Its disclosed base is

```
#print axioms lambda_unique_under_block = [A6'_block_
consistent, propext, Quot.sound, Classical.choice]
```

— one declared, non-core project axiom *plus the Lean core*.

Theorem 4.4 was honest but carried a cost: the trusted base included a *project axiom* that could not itself be discharged in-kernel. v24 removes that cost.

4.4 The v24 advance: axiom-free conditional uniqueness (CUT-2)

The v24 headline replaces the declared A6′ idealization with a *checkable structural property* of the candidate aggregator: *slice-multiplicativity* (separability). Informally, Φ is slice-multiplicative if it factors as a product of per-axis slice functions, each of which is multiplicative, normalized, and monotone — properties one can in principle *verify* of a concrete Φ , rather than postulate.

Definition 4.5 (Slice-multiplicativity / separability). $\Phi : [0, 1]^k \rightarrow [0, 1]$ is *slice-multiplicative* if there exist slice functions $f_i : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that (sep) $\Phi(x) = \prod_i f_i(x_i)$, (mul) $f_i(st) = f_i(s) f_i(t)$, (one) $f_i(1) = 1$, and (mono) each f_i is monotone.

Theorem 4.6 (Axiom-free conditional uniqueness of Λ — the v24 advance). [EXPERIMENTAL, AXIOM-FREE, CI-GREEN] Let $k > 0$ and let Φ satisfy the Lutar axioms $\{A1, A2, A3, A5\}$. If Φ is slice-multiplicative (Definition 4.5) then $\Phi = \Lambda_k$. The Lean term is *lambda_unique_of_separable* (namespace *Lutar.Round13*); its trusted base is exactly

```
#print axioms lambda_unique_of_separable = {propext, Classical.choice, Quot.
sound},
```

i.e. no project axiom. It is kernel-clean, CI-green, and present at main *ba1050b7* (originally merged at *b910c276*).

The exact Lean signature is:

```
theorem lambda_unique_of_separable {k : ℕ} (hk : 0 < k)
  (Φ : Aggregator k) (hL : LutarAxioms Φ)
  (f : Fin k → (NNReal → NNReal))
  (hsep : ∀ x, Φ x = ∏ i, f i (x i))
  (hmul : ∀ i s t, f i (s * t) = f i s * f i t)
  (hone : ∀ i, f i 1 = 1)
  (hmono : ∀ i, Monotone (f i)) :
  Φ = Λ k
```


Proof sketch and the in-tree lemma chain. The proof reduces slice-multiplicativity to factorization and then discharges via Theorem 4.3. Concretely:

1. **Per-axis power law.** For each axis i , the slice f_i is multiplicative (mul), monotone (mono), and normalized (one). The in-tree lemma `multiplicative_monotone_isPow_pos` yields an exponent α_i with $f_i(t) = t^{\alpha_i}$ for all $t \neq 0$:

```
multiplicative_monotone_isPow_pos {f} (hf_mul) (hf_mono) (hf_one)
  :  $\exists \alpha, \forall t \neq 0, f\ t = t^{\alpha} (\alpha : \mathbb{R})$ 
```

Boundary value $f_i(0) = 0^{\alpha_i}$ follows from the idempotent dichotomy (`slice_zero_idem`, `slice_const_one_of_zero_one`).

2. **Exponent equality across axes.** The chosen exponents α_i are forced equal by permutation invariance (A5): applying `Equiv.swap i j` to a two-hot test vector and using the in-tree injectivity lemma `rpow_left_inj_one_lt` ($c^a = c^b \Rightarrow a = b$ for $c > 1$) gives $\alpha_i = \alpha_j$.
3. **Assemble factorization.** Collecting the per-axis power laws yields the `Factors` predicate ($\Phi(x) = \prod_i x_i^{\alpha_i}$).
4. **Discharge.** Apply `lambda_unique_of_factors` (Theorem 4.3); idempotence (A3) and symmetry (A5) collapse the exponents to $1/k$ and conclude $\Phi = \Lambda_k$.

The whole chain `multiplicative_monotone_isPow_pos` $\rightarrow$ `lambda_unique_of_factors` discharges only through Mathlib and already-verified, sorry-free Round-13 lemmas; no new axiom token is introduced. Hence the trusted base is exactly the Lean core `{propext, Classical.choice, Quot.sound}`. $\square$

Remark 4.7 (Why this is strictly stronger than v23). Slice-multiplicativity is logically *weaker* than the *Factors* premise of Theorem 4.3 (it does not pre-suppose the explicit exponent vector; the exponents are *derived* via the power-law lemma), and it is a *checkable property of Φ* rather than a declared axiom about aggregation. Theorem 4.6 therefore (a) widens the gap between premise and conclusion and (b) removes the project axiom from the trusted base that Theorem 4.4 required. This is the sense in which v24 *strictly strengthens* the v23 conditional uniqueness result.

4.5 Λ is Conjecture 1 — the precise claim structure

We separate three propositions and assert the truth-value of each:

(U) **Unconditional uniqueness** “ Λ is the unique Φ satisfying A1–A5.” — **FALSE** (Theorem 4.2).

(C) **Conditional uniqueness** “Given $\{A1, A2, A3, A5\} +$ slice-multiplicativity, Λ is unique.” — **PROVEN, conditional, axiom-free** (Theorem 4.6).

(Conj) **The governance conjecture** “The *right* axiomatization of a governed trust aggregator forces Λ .” — **Conjecture 1**: a research hypothesis, not a theorem.

Conjecture 1 (The Lutar invariant; never a theorem). [CONJECTURE 1 – NOT A THEOREM]
The equal-weight geometric mean Λ_k is the correct unique trust aggregator for governed AI, in the sense that the governance-correct structural demand (slice-multiplicativity / aggregation-invariance) is the right normative requirement on a governed aggregator. This is an open philosophical claim, not a mathematical theorem; it is never asserted as proven, and the unconditional form (U) is provably false.

4.6 Worked examples

- **Disqualifying-zero.** $\Lambda(0.9, 0.9, 0.0) = 0$ — a single failed axis vetoes the verdict, unlike the arithmetic mean (0.6). Trust is not bought back by strong performance elsewhere.
- **Scale-covariance (A2).** $\Lambda(2x) = 2\Lambda(x)$: rescaling the evidence rescales the verdict, the measurement-theoretic objectivity property.
- **maxAgg comparison.** At (0.8, 0.2): $\Lambda = 0.4$ while $\text{maxAgg} = 0.8$ — the discriminating instance witnessing Theorem 4.2.
- **Slice-multiplicativity holds for Λ .** Take $f_i(t) = t^{1/k}$: then $\prod_i f_i(x_i) = (\prod_i x_i)^{1/k} = \Lambda_k(x)$, each f_i is multiplicative, $f_i(1) = 1$, and monotone — so the hypothesis of Theorem 4.6 is realized by Λ itself (non-vacuity).

4.7 CUT-1: the Aczél quasi-arithmetic representation theorem, closed on its stated hypotheses

Theorem 4.6 discharges slice-multiplicativity through a per-axis power-law step. That step is the shadow of a deeper classical result — the *Aczél quasi-arithmetic representation theorem*: a sufficiently regular bisymmetric mean F admits a continuous strictly-monotone generator φ with $F(x, y) = \varphi^{-1}((\varphi x + \varphi y)/2)$ [1, 2, 8, 9]. Internally we call the formalization of this representation **CUT-1**. The v24.1 advance is that **CUT-1 is now fully closed on its stated, checkable hypotheses**, kernel-clean, across Waves 18–22. We state the result, the construction, and — with care — what “closed on its stated hypotheses” does and does not mean.

What is now proven

Theorem 4.8 (CUT-1: quasi-arithmetic representation, forward direction). *[EXPERIMENTAL, AXIOM-FREE, CI-GREEN] Let F be reflexive, symmetric, partially strictly monotone, and quasi-arithmetic (admitting strictly-monotone φ, ψ with the dyadic midpoint recursion $F(fa)(fb) = f((a + b)/2)$, ψ continuous). Then the BKS dyadic generator f has dense range and extends to a continuous strictly monotone generator — the conclusion of the BKS forward construction. Every declaration in the Wave-18–22 chain has `#print axioms` $\subseteq \{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$, with no `sorry` and no new axiom token.*

The classical forward proof (Aczél 1948/1966 ch. 6; Burai–Kiss–Szokol [8, 9]) decomposes into a dyadic recursion, a strict-monotonicity / well-definedness step, an additive “generator collapse”, a *density of the dyadic image* step, a continuous extension, and the homogeneity endpoint that pins the geometric mean. Each is now discharged kernel-clean:

Step	Wave	What closed it (kernel-clean)
Forward fragment	18	AczelRepresentation/Cut1Chain: soundness of the quasi-arithmetic axioms; the analytic generator-collapse (<code>generator_unique_up_to_affine</code> , a monotone-additive rational squeeze); the <code>expMidpoint</code> witness pinning the geometric mean.
Density engine	19	DisjointOpens/Density: the BKS Step-2 contradiction — “uncountably many pairwise-disjoint open intervals cannot live on a separable line” (<code>false_of_uncountable_pairwiseDisjoint_Ioo</code>).
Primitives	20	countable-disjoint-opens and $\text{perfect} \Rightarrow \text{uncountable}$ bridges (standalone, also present in Mathlib).
(B) residual	21	the perfect-set obligation <i>avoided</i> : a strictly monotone generator is injective, so its image is a continuum and the accumulation set is uncountable (<code>range_not_countable_of_strictMono</code>) — the BKS parent-paper Theorem 8 <i>First step</i> , no Cantor–Bendixson.
(C-order) residual	22	the BKS <i>Fourth-step</i> gap-shift ordering $R s \leq L t$ for $s < t$ — previously carried as a stated hypothesis — is now <i>derived</i> from the discrete midpoint chain + monotone limit + continuity of ψ (<code>corder_gapshift</code> , <code>corder_data</code>).

The assembled chain is

$$\begin{array}{c}
 \text{StrictMono } f \quad + \quad \text{quasi-arithmetic} \quad + \quad \psi \quad \text{continuous} \quad \xrightarrow{\text{Wave-22 derives (C-order)}} \quad \text{the} \\
 \text{exact hC of dyadic_image_dense_complete} \quad \xrightarrow{\text{Wave-21 ((B) internal)}} \quad \text{DenseRange } f \\
 \xrightarrow{\text{Wave-18 = Mathlib Monotone.continuous_of_denseRange}} \text{Continuous } f.
 \end{array}$$

The topological density engine (the disjoint-opens contradiction, the gap extraction) was closed in Waves 19/21; Wave 22 closed the last residual (C-order). **Every residual is now closed on the stated, checkable hypotheses** (bisymmetry / quasi-arithmetic structure + partial strict monotonicity + reflexivity + symmetry — all *properties*, no axioms).

How CUT-1 sharpens the conditional Λ result

Closing CUT-1 lets us *drop* hypotheses from the conditional Λ -uniqueness theorem of §4.4. The previous frontier (`cut1_conditional_lambda`) carried five slice hypotheses; Wave 22 proves two of them redundant:

- `bisymmetry_is_redundant` — the slice-induced operation $(s, t) \mapsto f(s \cdot t)$ is bisymmetric whenever the slice is multiplicative, so slice-bisymmetry is a theorem, not an assumption.
- `slice_one_eq_one_of_sep` — unit-normalization $f_i(1) = 1$ is *derivable* from A3 diagonal normalization lifted through separability plus multiplicative idempotency, so it need not be assumed.

Theorem 4.9 (Sharpest conditional Λ -uniqueness). *[EXPERIMENTAL, AXIOM-FREE, CI-GREEN]*
Under $\{A1\text{--}A5\}$ together with separability ($\Phi x = \prod_i f_i(x_i)$), slice-multiplicativity ($f_i(st) =$

$f_i(s)f_i(t)$), and *slice-monotonicity* (f_i monotone), $\Phi = \Lambda_k$. The Lean term is `cut1_sharp_conditional_lambda`; it discharges axiom-free through the in-tree `lambda_unique_of_separable` (`#print axioms = {propext, Classical.choice, Quot.sound}`). This is strictly weaker than the prior *bisymmetric frontier* (two fewer hypotheses).

Why this is the sharp boundary. Slice-multiplicativity is the irreducible Cauchy-type input the false unconditional statement lacks; dropping it re-admits the `maxAgg/min` counterexamples and makes the conclusion FALSE. So the hypothesis set cannot be weakened further within the separable family.

What “closed on its stated hypotheses” means — precisely

This is the claim most easily misread, so we are explicit.

- **What CUT-1 closure *is*:** the quasi-arithmetic *representation* theorem (Theorem 4.8) is complete on its stated, checkable hypotheses — the BKS forward construction is built end to end, with the (B) and (C-order) residuals *derived*, not assumed and not axiomatised. It *sharpens* the conditional Λ -uniqueness theorem (Theorem 4.9) to its weakest checkable hypothesis set.
- **What CUT-1 closure *is not*:** it does *not* make Λ unconditional. The closure is for the quasi-arithmetic class — precisely the class the conditional- Λ chain inhabits — and it presupposes that structure. The *unconditional* uniqueness of Λ under {A1–A5} alone stays **Conjecture 1** and is machine-checked **false** (Theorem 4.2, `maxAgg_ne_Lambda`); that counterexample is untouched. CUT-1 strengthens a *conditional* result; it does not, and cannot, discharge the false unconditional one.
- **Tier.** All Wave-18–22 work is *experimental*, *axiom-free*, *CI-green* and is never folded into the locked-five.

Sources

The construction follows Burai–Kiss–Szokol [8] (Theorem 8, Fourth-step eqs. (8)–(9)), [9] (Lemma 6, Step 2), the n -ary regularity-free treatment [21], and Aczél–Dhombres [3] pp. 287–290. No source code is imported; all cited references are mathematical literature.

5 The proof-trail / receipt architecture

Every governed action emits a *receipt*: a canonical record of the action, its inputs, the gate decision, and the Λ sub-scores. The chain entry is $\text{link}_i = H(\text{payload}_i \parallel \text{link}_{i-1})$, and a batch commits to a Merkle root.

5.1 Structural correctness (sorry-free)

The replay engine (F1) is deterministic: equal logs replay to equal final state, and the folded state equals the last entry of the explicit replay trace (induction over the log; `#print axioms` reports `propext` only). Hash-chain verification is sound: a verified chain implies all links match. The Merkle inclusion checker is correct-by-construction over an abstract hash. A worked tamper check: altering any payload_i changes link_i , which propagates to root, so re-verification fails.

5.2 Security under declared idealizations (axiom-gated)

Tamper-evidence, attribution, and inclusion-binding require collision-resistance / unforgeability assumptions. We declare these as named axioms and disclose them, following the ProVerif methodology of abstracting a primitive as an axiom, proving the protocol against it, and discharging the axiom separately for a concrete implementation [6]. The declared axioms include `sha256`, `sha256_collision_resistant`, `node_collision_resistant`, `leaf_collision_resistant`, `compression_collision_resistant`, and `domain_separation`. Each is an injective-oracle abstraction — *not* a proof of cryptographic hardness — disclosed in the `#print axioms` ledger. The architecture retains the full sub-score vector (never letting Λ replace it), gates deny-by-default, and supports byte-identical deterministic replay (F1).

6 The locked kernel and the frontier theorem families

6.1 The locked five (kernel-verified at c7c0ba17)

The Lean formalization has a *locked kernel*, pinned at commit `c7c0ba17`: **749 declarations / 14 unique axioms / 163 sorries**. Within this counted scope, exactly **five** governance formulas are proven and locked.

ID	Formula	Status note
F1	Replay determinism	locked
F11	Ayni reciprocity	locked
F12	Kuramoto additive	locked (<i>additive scaffolding only</i> , not the full sync theorem)
F18	Reed–Solomon	locked
F19	Bekenstein additive	locked (<i>additive scaffolding only</i> , NOT the bound $S \leq 2\pi kRE/\hbar c$)

All other formal work — the experimental pass and every frontier family below — lives in a *separate experimental tier* and is **never** folded into the locked-five.

6.2 The experimental CI-green tier (at main ba1050b7)

At `main` commit `ba1050b7` (Waves 11–22 all merged, CI-green, drift-clean), the experimental tier comprises **~185 kernel-clean theorems**. The drift-gated baseline (computed over the locked scope, with Waves 15–22 registered under `EXPERIMENTAL_SCOPES` so they are additive) is **1323 declarations / 23 axioms (22 unique) / 307 sorries**. The frontier theorems below are all `#print axioms -clean` ($\subseteq \{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$), introduce no new axiom token, and add no `sorry`.

6.3 CF-13 — DEQ input-Lipschitz well-posedness margin

[[EXPERIMENTAL](#), [AXIOM-FREE](#), [CI-GREEN](#)] File `Lutar/Innovations/round5/OuroLoopInputLipschitz.lean` (7 declarations). For a contraction with input-Lipschitz constant L_x and contraction modulus $K < 1$, the equilibrium map z^* satisfies

$$\text{dist}(z^*(x), z^*(y)) \leq \frac{L_x}{1 - K} \text{dist}(x, y)$$

(`equilibrium_dist_le`, `equilibrium_lipschitz`). Built on Mathlib `ContractingWith`; the result is the well-posedness margin of a deep equilibrium model (Bai–Kolter–Koltun [5]).
`#print axioms = {propext, Classical.choice, Quot.sound}`.

6.4 CF-17 — floating-point summation error bound

[EXPERIMENTAL, AXIOM-FREE, CI-GREEN] File Lutar/Khipu/NumericStability.lean (8 declarations). Under the standard rounding model $\text{fl}(a + b) = (a + b)(1 + \delta)$, $|\delta| \leq u$ (carried as an explicit hypothesis, not an axiom; Higham [16] §2.2), the recursive sum satisfies $|\text{recSum}(x, \delta) - \sum_i x_i| \leq ((1 + u)^{n-1} - 1) \sum_i |x_i|$ (`recSum_error_le`). `#print axioms = {propext, Classical.choice, Quot.sound}`.

6.5 Wave-13 — replay completeness, quorum shadow, HM bottleneck

[EXPERIMENTAL, AXIOM-FREE, CI-GREEN] (File Lutar/Wave13/Sweep.lean and an in-place `close`.)

- `findReplayRoot_complete` (PRNG, `-1` sorry; closed in place via Lean core `List.find?_isSome`): if a replay-root candidate is a member of the candidate list and validates, the search returns `some`. `#print axioms = {propext, Quot.sound}`.
- `quorum_agreement_single_valued_vote`: for $n \geq 3f + 1$ and two quorums of size $\geq n - f$, single-valued votes agree ($v_1 = v_2$), by nonempty quorum intersection. `#print axioms = {propext, Classical.choice, Quot.sound}`. **Honesty**: this assumes each organ votes via a *total function* `voteOf`, so it is an explicitly-labeled *non-Byzantine shadow*, **not** Khipu Conjecture 2 (a faulty organ that equivocates).
- `hm_bottleneck_clean` (HLP harmonic-mean bottleneck): if the harmonic mean falls below a threshold then some coordinate does (contrapositive; pure order/field reasoning). `#print axioms = {propext, Classical.choice, Quot.sound}`.

6.6 Wave-14 — CF-18/19/20/21 (nine kernel-clean theorems)

[EXPERIMENTAL, AXIOM-FREE, CI-GREEN] All nine compile clean with `#print axioms = {propext, Classical.choice, Quot.sound}`.

- **CF-18 — Mādhava/Leibniz alternating-series remainder** (Lutar/Wave14/LeibnizRemainder.lean). For an antitone nonnegative a with alternating partial sums tending to L , one has $|\sum_{i < N} (-1)^i a_i - L| \leq a_N$ (the Lean terms `leibniz_remainder_bound` and `madhava_alt_series_bound_clean`), via Mathlib `Antitone.alternating_series_le_tendsto` even/odd bracketing (Plofker [25]).
- **CF-19 — Reed–Solomon MDS distance lower bound** (Lutar/Wave14/ReedSolomonDistance.lean). Two distinct polynomials of degree $< k$ agree on fewer than k points (`agreement_card_lt_of_degree_lt`); lifting via injectivity of the evaluation points gives the bound $n - k + 1 \leq \#\{i : p(\text{pts}_i) \neq q(\text{pts}_i)\}$ (`rs_distance_lower_bound`). This is the *achievability / lower-distance half* of Singleton; the upper bound and full MDS equality remain honest sorries (Reed–Solomon [26], Singleton [27]).
- **CF-20 — VCG efficiency + truthfulness core** (Lutar/Wave14/VCGEfficiency.lean). An efficient outcome exists and maximizes social welfare (`exists_efficient_outcome`,

efficientOutcome_maximises); the truthfulness core is the marginal-contribution monotonicity consequence (vcg_truthfulness_core) (Vickrey–Clarke–Groves [29, 10, 15]; Nisan et al. [23]). **Honesty:** the in-tree MechanismDesign/VCG.lean references a nonexistent Finset.argmax and is unwired; CF-20 is a clean companion, not a patch.

- **CF-21 — Cover–Thomas log-sum + Gibbs** (Lutar/Wave14/LogSumInequality.lean). The log-sum inequality (log_sum_inequality) and Gibbs’ inequality (gibbs_inequality) via $\log x \leq x - 1$ (Cover–Thomas [11] Thm 2.7.1 and 2.6.3). **Honesty:** this is the *correctly-stated* DPI core; it does **not** repair the in-tree DPO klDivergence/pinsker, which remain false-as-stated for lack of a simplex hypothesis.

6.7 Wave-15 — CF-22: simplex-conditional DPO KL repair

[EXPERIMENTAL, AXIOM-FREE, CI-GREEN] (Lutar/Wave15/.) The in-tree DPO axiom klDivergence_nonneg is *false as stated* (no simplex constraint). CF-22 (klDivergence_nonneg_simplex, dpo_klDivergence_nonneg_on_simplex) proves the *correct* statement: KL divergence is nonnegative *on the probability simplex*, axiom-free, via Gibbs’ inequality. **Honesty:** this *conditionally repairs* the claim on the simplex; the unconditional DPO axiom stays false-as-stated and untouched. #print axioms = {propext, Classical.choice, Quot.sound}.

6.8 Wave-16/17 — CF-23 full binary Pinsker; CF-24–28

[EXPERIMENTAL, AXIOM-FREE, CI-GREEN]

- **CF-23 — the full binary Pinsker inequality** (Wave-17, binary_pinsker): $2(p - q)^2 \leq \text{KL}(p \parallel q)$ for the binary distribution — the long-sought headline, built on the binary-KL crux binary_inv_sum_ge_four (Wave-16). #print axioms = {propext, Classical.choice, Quot.sound}.
- **CF-24 — geoBin satisfies the full Aczél quasi-arithmetic axioms** (Wave-16): idempotence, commutativity, homogeneity, monotonicity — the first real CUT-1 progress and the bridge to §4.7.
- **CF-25 — Λ scale-invariance** (Wave-16, lambda_scale_axes, normalization_invariant).
- **CF-26 — abacus place-value** (Wave-16): a positional place-value lemma.
- **CF-27 — monotone-DEQ unique equilibrium** (Wave-17): strong monotonicity $\Rightarrow$ a unique equilibrium.
- **CF-28 — recurrent-depth Lipschitz contraction** (Wave-17): a K^r -Lipschitz contraction-amplification bound for recurrent depth (after the retrofitting-recurrence line, Apache-2.0).

All Wave-16/17 declarations are #print axioms -clean ($\subseteq$ {propext, Classical.choice, Quot.sound}).

6.9 Waves 18–22 — the CUT-1 construction

[EXPERIMENTAL, AXIOM-FREE, CI-GREEN] The Wave-18–22 CUT-1 chain (the Aczél quasi-arithmetic representation theorem, closed on its stated hypotheses, plus the sharpened conditional Λ theorem) is treated in full in §4.7. Every one of its declarations is #print axioms -clean ($\subseteq$ {propext, Classical.choice, Quot.sound}), with no sorry and no new axiom token.

7 Verification status, in detail

7.1 The wave ledger (Waves 11–22)

Wave	PR	Adds	Headline
11	#201	24 thm	CF-1/2/3/5 position-aware graph + KV-cache + early-exit + Neyman–Pearson
12	#202	CUT-2 + CF-13 + CF-17	axiom-free conditional Λ (<code>lambda_unique_of_separable</code>); DEQ; fp-summation; -1 legacy sorry
13	#203	3 thm	replay-root completeness (-1 sorry); non-Byzantine quorum shadow; HM bottleneck
14	#204	9 thm	CF-18 Mādhava/Leibniz; CF-19 RS lower bound; CF-20 VCG; CF-21 log-sum/Gibbs
15	#205	CF-22 +	CF-22 simplex-conditional DPO KL repair; CUT-1 bisymmetry bridge
16	#206	13 thm	CF-23 binary-KL crux; CF-24 geoBin Aczél axioms; CF-25 Λ scale-inv.; CF-26 abacus
17	#207	24 thm	CF-23 full binary Pinsker ; CF-27 mon-DEQ uniqueness; CF-28 recurrent-depth Lipschitz
18	#208	19 thm	CUT-1 forward fragment (Aczél representation; generator-collapse)
19	#209	21 thm	CUT-1 BKS density engine (disjoint-opens contradiction)
20	#210	17 thm	CUT-1 primitives (countable-disjoint-opens; perfect $\Rightarrow$ uncountable)
21	#211	13 thm	CUT-1 (B) residual via monotone-extension uncountability
22	#212	15 thm	CUT-1 (C-order) derived $\Rightarrow$ closed on stated hyps; <code>cut1_sharp_conditional_lambda</code>

All Waves 11–22 are merged to `main` at `ba1050b7` (CI-green), totalling ~ 185 kernel-clean theorems in the experimental tier.

7.2 The drift gate and the canonical numbers

A numbers-drift gate guards four quantities and the axiom-name set: *declarations*, *axioms_raw*, *axioms_unique*, and *sorries_raw*. At `main ba1050b7` the canonical figures are **1323 / 23 / 22 / 307** — *unchanged* since `b910c276`: every Wave-15–22 module is registered under `EXPERIMENTAL_SCOPES`, so its declarations are additive and do not move the locked baseline. The gate is green and drift-clean across all of Waves 11–22.

7.3 The `#print axioms` discipline

Every claimed-proven theorem is accompanied by its `#print axioms` ledger. The headline theorem and every frontier family member report `#print axioms` $\subseteq$ `{propext, Classical.choice, Quot.sound}` — the trusted Lean core only — with the sole exception of the explicitly axiom-gated cryptographic results (§5), whose declared idealizations appear in their ledgers. No theorem is stated as proven that CI has not verified at the recorded commit.

7.4 SLSA supply-chain posture — correcting a v23 overstatement

v23 wrote “SLSA L1+L2 attested”. **This badge was wrong and is corrected here.** The honest posture is **SLSA L1, with an L2 build-attestation present** on organ images. We do *not* claim *L2-verified*, L3, FedRAMP authorization, Iron Bank accreditation, or CMMC certification [24]; those remain roadmap. The container build pipeline meets L1 provenance expectations and carries an L2-style build-provenance attestation, but full L2 *verification* is not yet earned and is reported as a roadmap item rather than a current state.

8 Why this is groundbreaking — the honest case

Axiom elimination as a first-class result. The standard way to “strengthen” a conditional theorem is to weaken its hypothesis. v24 does something sharper: it *removes a project axiom from the trusted base entirely*. v23’s conditional uniqueness depended on `A6'_block_consistent`, a declared idealization that the kernel had to *trust*. v24’s `lambda_unique_of_separable` depends on no project axiom at all (`#print axioms = {propext, Classical.choice, Quot.sound}`); the only added hypothesis is a *checkable property of Φ* . To our knowledge, exhibiting a governed-AI aggregator uniqueness result whose trusted base is exactly the bare Lean core is new.

A machine-checked boundary, not just a conditional. We did not merely prove a conditional theorem; we proved the *negation* of the unconditional claim (Theorem 4.2). Establishing exactly which extra assumption is required — and that without it the claim fails — converts a routine conditional into a *boundary* result: Theorem 4.6 is demonstrably close to the maximal true statement of its kind, and the boundary itself is machine-checked.

Disclosed-axiom discipline at scale. Formal verification of security-critical systems is well established (seL4 is the canonical machine-checked OS kernel [17]); applying a disclosed-axiom, kernel-checked discipline to an AI governance aggregator and its receipt architecture — with a `#print axioms` ledger for *every* claim and a hard separation between a locked five and an experimental tier — is, as far as we can determine, new.

8.1 Honest comparison to prior art

Prior art	What it provides	How v24 differs (honestly)
Verifiable-claims governance [7, 22]	The argument that AI claims should be externally verifiable.	A concrete substrate with a kernel-checked core; does not claim to solve verifiability in general.
seL4 / formal OS [17]	Machine-checked OS-kernel correctness; the gold standard.	Verifies a governance aggregator + receipt layer; far smaller in scope and maturity; no seL4-level coverage claimed.
Certificate Trans- parency [19]	Merkle-based tamper-evident logs in production.	Reuses the Merkle discipline; formalizes the binding property in Lean under declared axioms.
Functional-equation aggregation [1, 4, 12]	Uniqueness characterizations of the geometric mean.	Formalizes an <i>axiom-free</i> conditional uniqueness (slice-multiplicativity) and machine-checks the unconditional refutation.
SLSA framework [24]	A supply-chain integrity ladder (L1–L4).	Honestly attests L1; L2 is a roadmap item; L3 / FedRAMP / Iron Bank / CMMC explicitly <i>not</i> claimed.

Honest summary. The individual mathematical facts (geometric-mean characterizations, Kraft, Shannon, Byzantine quorums, Reed–Solomon, VCG, log-sum) are classical. The contribution of v24 is their unification into a governed-AI trust substrate, their kernel-checked instantiation, and above all the *axiom elimination* on the central uniqueness result combined with the *honesty architecture* — locked counts, disclosed axioms, and a machine-checked refutation guarding the one tempting overclaim.

9 Philosophical foundations: slice-multiplicativity as an epistemic upgrade

9.1 From a declared axiom to a checkable property

A trusted base that contains a project axiom asks the auditor to *believe* something the kernel cannot check. A6' (block-consistency) was such an axiom: governance-legible, published, and defensible, but ultimately *postulated*. Slice-multiplicativity is different in kind. It is a structural *property of the candidate aggregator* Φ — of the form “ Φ factors into multiplicative, normalized, monotone slices” — that one can in principle *verify* of a concrete Φ by inspecting its definition. Replacing a postulate the auditor must trust with a property the auditor can check is, on any reliabilist account of justification [14], an *epistemic upgrade*: the warrant for the conclusion no longer routes through an unverifiable idealization.

9.2 When is a conditional legitimate?

Four conditions, all met by Theorem 4.6: (i) *independently-motivated antecedent* — separability is the standard hypothesis under which multiplicative aggregators are characterized [2]; (ii) *non-vacuous instantiation* — Λ itself is slice-multiplicative (§4.6), so the conditional is not vacuously true; (iii) *boundary established* — without the hypothesis, uniqueness fails (Theorem 4.2); (iv)

disclosed at point of use — the `#print axioms` ledger shows the bare core. The residual gap — *is slice-multiplicativity the philosophically correct demand on a governed aggregator?* — is genuinely open, which is exactly why Λ remains Conjecture 1.

9.3 Slice-multiplicativity vs. block-consistency

Both $A6'$ and slice-multiplicativity are sufficient to pin Λ . The relevant difference is epistemic, not merely logical. $A6'$ entered the trusted base as a named axiom; slice-multiplicativity enters as a *hypothesis discharged through the kernel*. The former widens the trusted computing base; the latter does not. We therefore regard the v24 route as dominating the v23 route along the one axis that matters to an auditor: *how much must be trusted that cannot be checked*.

9.4 Steelman objections (with honest rebuttals)

1. “*Slice-multiplicativity is the geometric mean in costume.*” — It is strictly weaker than the explicit *Factors* premise and does not name the exponents; the power-law lemma *derives* them. The residue is a defensible normative choice, labeled as Conjecture 1.
2. “*Motte-and-bailey.*” — The conditional theorem and the bald “ Λ uniqueness” are labeled and separated in the artifacts themselves; (U) is stated false.
3. “*Idealized crypto axioms make guarantees illusory.*” — Conceded and disclosed; receipt guarantees are conditional on the same assumptions TLS and Certificate Transparency already rely on [19].
4. “*Arrow: aggregation is impossible.*” — Arrow concerns ordinal preference aggregation; Λ aggregates cardinal ratio-scale evidence about one object, outside Arrow’s IIA impossibility.
5. “*Gödel/Tarski: self-auditing is incoherent.*” — Conceded as a limit; the substrate uses bounded recursion verified by an external stratified stack, not self-verification.

10 Empirical posture (measured, not proven)

The figures below are *measured*, reported as empirical percentiles over archived runs, with no theorem claimed. Per-receipt build latency $\approx 11.5 \mu s$ (p50) / $50.7 \mu s$ (p99); verification $\approx 10.4 \mu s$ (p50). Nine-axis $\Lambda \approx 3.12 \mu s$ (base) / $3.29 \mu s$ (composed). On a prior platform run of 24,800 HTTP calls, governance overhead was 0.49–0.59 ms per route (p50), 1.27 ms (p99). A deterministic 5× replay yields a byte-identical root.

Operation	p50	p99	What is measured
Receipt build	$11.5 \mu s$	$50.7 \mu s$	Canonicalize + hash + chain + persist
Receipt verify	$10.4 \mu s$	—	Chain re-check + root comparison
Λ_9 (base)	$3.12 \mu s$	—	Nine-axis geometric mean, isolated
Λ_9 (comp.)	$3.29 \mu s$	—	Within a composed gate evaluation
Gov. overhead / route	0.49–0.59 ms	1.27 ms	Governed minus ungoverned, 24,800 calls

These are wall-clock timings collected in-process with a monotonic nanosecond clock over archived runs; no confidence intervals or cross-run variance are claimed.

11 Limitations and honesty

- **Λ is Conjecture 1 unconditionally.** Unconditional uniqueness under A1–A5 is false (Theorem 4.2). Only conditional theorems hold, and whether slice-multiplicativity is the philosophically correct demand is open. Λ is never claimed proven unconditionally. **CUT-1 closure (§4.7) sharpens the *conditional* result and does not change this:** the quasi-arithmetic representation is closed on its stated hypotheses, but the unconditional conjecture stays open and machine-checked false.
- **The locked count is exactly 5.** {F1,F11,F12,F18,F19} at c7c0ba17 (749/14/163). The experimental CI-green tier (~185 kernel-clean theorems; drift baseline 1323/23/22/307 at main ba1050b7) is separate and never folded in.
- **Byzantine BFT safety stays Khipu Conjecture 2 (open).** A faulty organ can equivocate; `ubuntu_quorum_safety` is untouched. The Wave-13 `quorum_agreement_single_valued_vote` is an explicitly non-Byzantine shadow, not a resolution.
- **DPO `klDivergence_nonneg` / `pinsker` are false as stated** (no simplex constraint on `PolicyParam`); they remain honest declared axioms. CF-21 proves the correctly-stated `logsum/Gibbs` core but does *not* repair them.
- **Open sorries remain.** The locked kernel carries 163 sorries; the experimental scope carries 307 raw sorries. RS Singleton upper bound / full MDS equality, the in-tree VCG file, Bekenstein/Bousso IQ-gate stubs, and the hard sites (Putnam analysis, xoshiro period via $\text{GF}(2)^{256}$ primitivity, Hoeffding–Azuma, Brouwer/cohomology, gauge symmetry, path-integral convergence) stay honest sorries with references intact.
- **SLSA L1 honest, L2 build-attestation present.** Not L2-verified, L3, FedRAMP, Iron Bank, or CMMC (§7.4).
- **Declared idealizations.** Tamper-evidence, attribution, and Merkle binding hold only under declared collision-resistance / unforgeability axioms, disclosed in every `#print axioms` ledger.
- **Scope.** The substrate verifies a governance aggregator and receipt layer, not an end-to-end AI system; it is far smaller in scope and maturity than a fully verified OS kernel.

12 Conclusion

v24 advanced the Ouroboros program by a single, sharply-stated step: the central trust aggregator’s conditional uniqueness, which in v23 rested on a declared block-consistency axiom, now rests on *no project axiom at all*. The in-tree `lambda_unique_of_separable` proves Λ -uniqueness under {A1,A2,A3,A5} plus slice-multiplicativity with `#print axioms = {propept, Classical.choice, Quot.sound}`, kernel-clean and CI-green at ba1050b7, and the extra hypothesis is a checkable property of Φ rather than an idealization. This revision (v24.1) adds the further advance that **CUT-1 — the Aczél quasi-arithmetic *representation* theorem — is now fully closed on its stated, checkable hypotheses** (Waves 18–22, §4.7), which in turn sharpens the *conditional* Λ -uniqueness chain to be axiom-clean end to end on {A1–A5} + separability + slice-multiplicativity + slice-monotonicity (`cut1_sharp_conditional_lambda`). The *unconditional* claim remains machine-checked *false*, so Λ stays Conjecture 1, exactly as before — CUT-1 does not, and cannot, change that.

The one-line thesis. *v24 strictly strengthens the v23 conditional uniqueness of the Lutar invariant by eliminating the declared block-consistency axiom: Λ -uniqueness is now proven conditional on $\{A1, A2, A3, A5\}$ plus a checkable slice-multiplicativity property with a trusted base of exactly the bare Lean core, while its unconditional uniqueness remains machine-checked false — so the axiom-free conditional theorem is the maximal true statement and the invariant is honestly labeled Conjecture 1, never a theorem.*

The contribution is not a new theorem alone but a new *standard*: axiom elimination on the load-bearing result, locked counts held separate from an experimental tier, disclosed axioms, and a machine-checked refutation guarding the one tempting overclaim. We submit that this calibrated honesty is precisely what a regulator, an auditor, or a defense customer should require.

A Reproducibility and artifact manifest

A.1 Pinned commits and artifacts

Component	Pin	Contents / status
main (Waves 11–22 merged)	ba1050b7	1323 decl / 23 axioms (22 unique) / 307 sorries; ~185 experimental kernel-clean thms; CI-green
Prior pin (Waves 11–14/15)	b910c276	superseded by ba1050b7; same drift baseline
Locked kernel	c7c0ba17	749 / 14 / 163; locks {F1,F11,F12,F18,F19}
CUT-2 axiom-free Λ	in ba1050b7	Round13/LambdaSeparable.lean
CUT-1 representation (Waves 18–22)	in ba1050b7	AczelRepresentation/; closed on stated hypotheses
Concept DOI (always-latest)	—	10.5281/zenodo.19944926
Repository	—	github.com/szl-holdings/lutar-lean

A.2 How to verify

```
# 1. Kernel-check the merged library at the pinned commit
$ git checkout ba1050b7 && lake build # expect: green, no errors

# 2. Disclose the trusted axiom base of the v24 headline theorem
$ #print axioms Lutar.Round13.lambda_unique_of_separable
=> [proptext, Classical.choice, Quot.sound] # NO project axiom

# 3. Confirm the unconditional claim is refuted
$ #print axioms maxAgg_ne_Lambda # Lean-core axioms only

# 4. Verify the locked kernel separately
$ git checkout c7c0ba17 && lake build # locks F1,F11,F12,F18,F19
```

A.3 Trusted computing base

The bottom turtles, stated once: (i) soundness of the Lean 4 kernel and its core axioms {proptext, Classical.choice, Quot.sound}[13, 28]; (ii) the declared cryptographic idealizations (sha256_collision_resistant and the Merkle collision-resistance / domain-separation axioms), used only in the receipt-security theorems; and (iii) human sign-off gating CI. The v24 headline theorem `lambda_unique_of_separable` depends on *none* of the cryptographic idealizations and on *no* project axiom — only on (i).

B Notation and glossary

Symbol / term	Meaning
$\Lambda_k(x)$	The Lutar invariant: equal-weight geometric mean $(\prod_i x_i)^{1/k}$ on $[0, 1]^k$
Φ	A generic candidate aggregator $[0, 1]^k \rightarrow [0, 1]$ satisfying some subset of A1–A5
maxAgg, min	The machine-checked A1–A5 witnesses that A1–A5 do not pin Λ (Thm 4.2)
A1–A5	Monotonicity, positive homogeneity, idempotence, boundedness, permutation-invariance
A6'	Block-consistency: the v23 <i>declared</i> axiom (replaced in v24)
slice-multiplicativity	The v24 <i>checkable</i> hypothesis: $\Phi = \prod_i f_i(x_i)$ with each f_i multiplicative, normalized, monotone
lambda_unique_of_separable	The v24 headline: axiom-free conditional Λ -uniqueness
link _i , root	The <i>i</i> -th hash-chain entry and the Merkle commitment to a receipt batch
sorry-free	A Lean development with no <code>sorry</code> ; the kernel checks every step
CI-green	The real CI <code>lake build</code> (a full kernel check) passes at the pinned commit
axiom-gated	Sorry-free given an explicitly declared, disclosed idealizing axiom
#print axioms	Lean command listing the trusted axiom base of a theorem
locked-five	The exactly five kernel-verified formulas {F1,F11,F12,F18,F19} at c7c0ba17
experimental tier	The separate CI-green corpus (~ 185 kernel-clean thms; drift 1323/23/307 at ba1050b7), never folded into the locked-five
Conjecture 1	The open claim that Λ is the correct unique governed aggregator; never a theorem

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Honesty doctrine preserved verbatim: Λ is Conjecture 1 unconditionally (never a theorem); the v24 conditional theorem `lambda_unique_of_separable` holds under `{A1,A2,A3,A5}` + slice-multiplicativity with `#print axioms = {propext, Classical.choice, Quot.sound}`(no project axiom); locked/proven = exactly five; the experimental CI-green tier is separate and never folded in; declared axioms disclosed in every `#print axioms` ledger; SLSA L1 honest with L2 build-attestation present (L2-verified, L3, FedRAMP, Iron Bank, and CMMC remain roadmap). No fabricated results, no fake citations, 0 runtime CDN.